北京交通大學 BEIJING JIAOTONG UNIVERSITY

Zhou Hang

BJTU Signals and Systems Course Group, Copyright © 2020

Continuous Time Signals

- **Regular signals**
  - DC signals
  - Sinusoidal signals
  - Exponential signals
  - Sampling signals
- **Singularity**
  - Unit step signal
  - Impulse signal
  - Ramp signals
  - Impulse pair signals

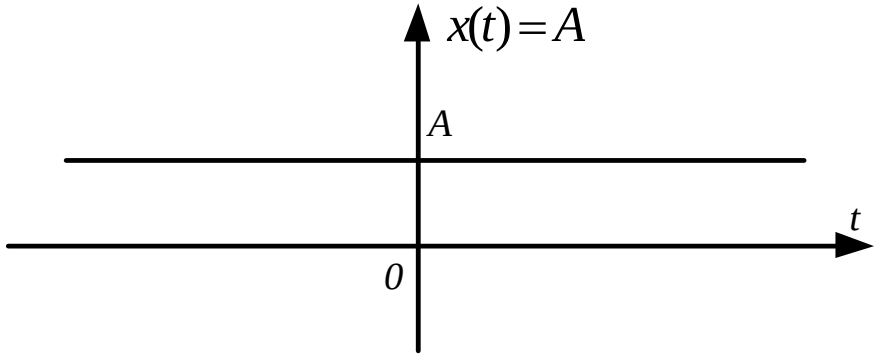
Regular signals

- Definition

the function $x(t)$ representing the signal has no discontinuities (jump points) in itself and its derivative $x'(t)$ or higher derivative $x^{(n)}(t)$

1. DC Signals

$$x(t) = A, \qquad -\infty < t < +\infty$$



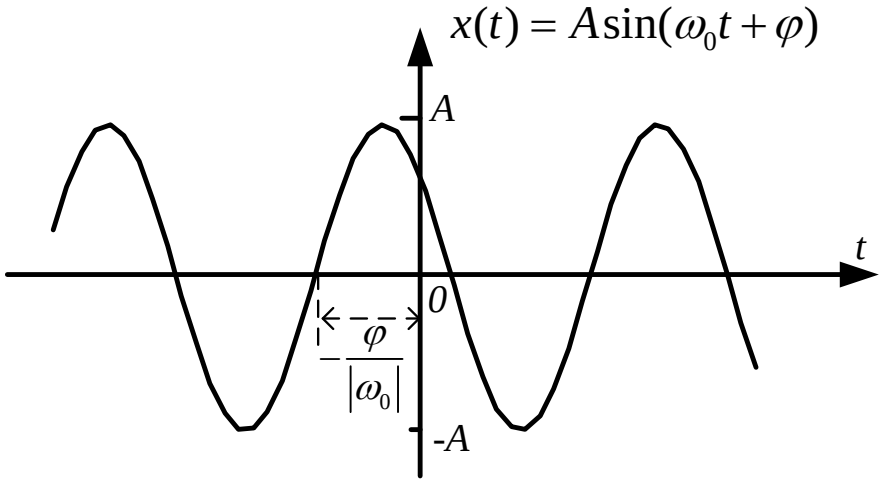
Battery

Signal source

2. Sinusoidal signals

$$x(t) = A\sin(\omega_0 t + \varphi)$$

$$x(t) = x(t + T_0)$$



period: $T_0 = \dfrac{2\pi}{|\omega_0|}$

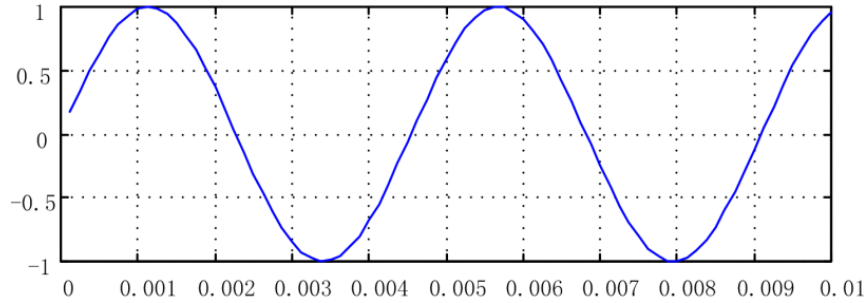
frequency: $f_0 = \dfrac{1}{T_0}$

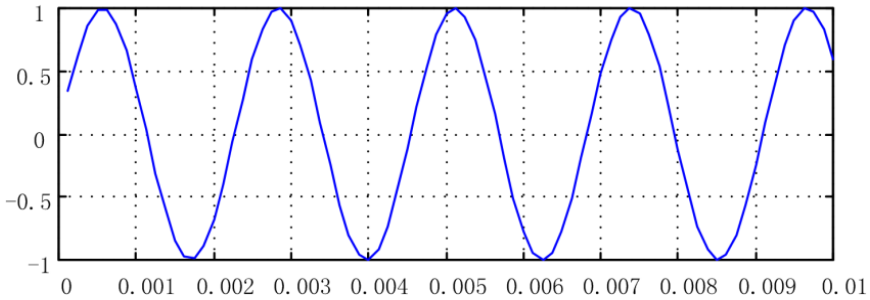
A(amplitude) ω_0 (angular frequency) $w_0 = 2\pi f_0$

φ (initial phase)

2. Sinusoidal signals

Waveform of sinusoidal signals $x_1(t) = \sin(2\pi \times 220t), x_2(t) = \sin(2\pi \times 440t)$



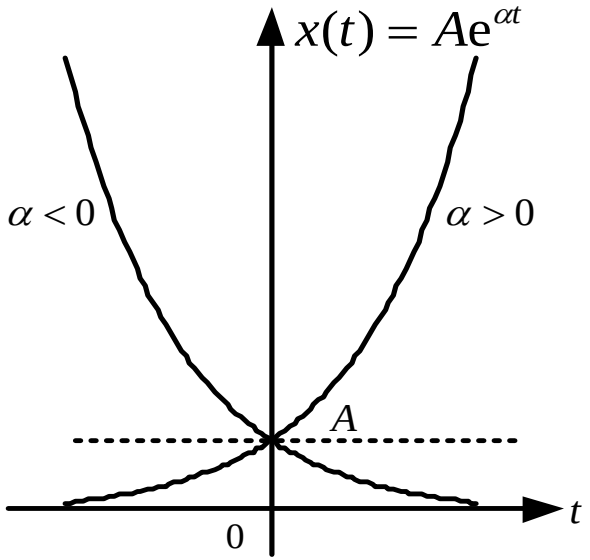


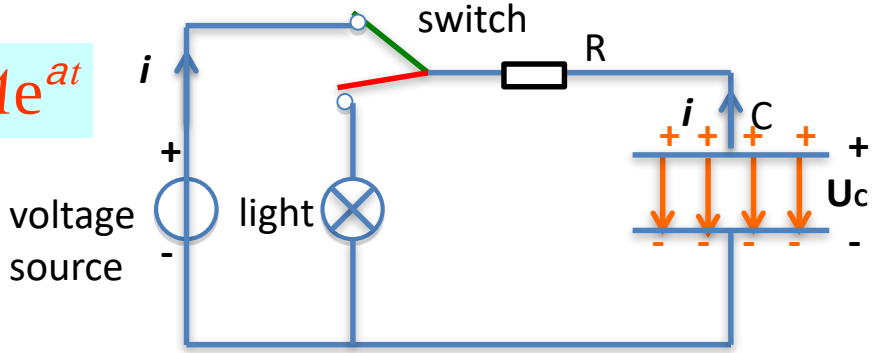
The higher the frequency, the faster the signal oscillates and the sharper the sound.

3. Exponential signals

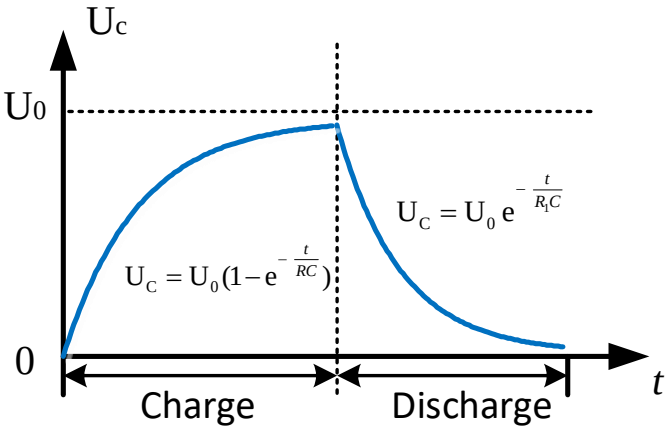
- Real exponential signals

$$x(t) = Ae^{at}$$



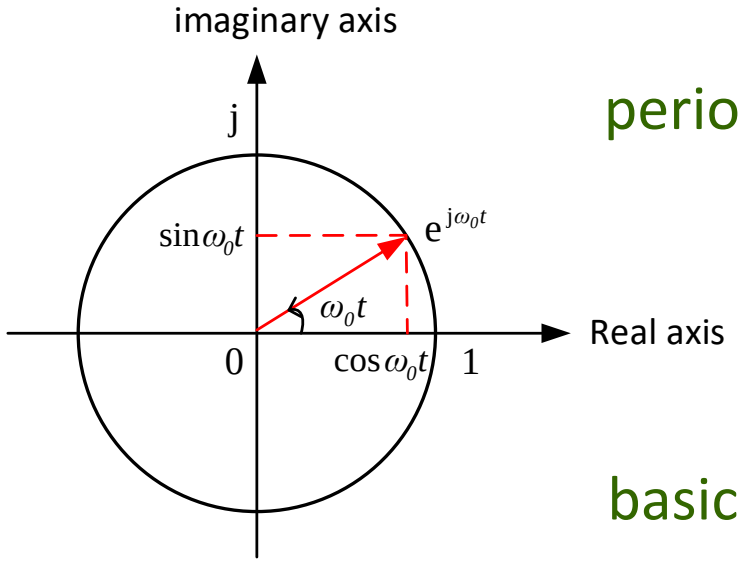


Capacitor charging and discharging process



3. Exponential signals

- Imaginary exponential signals $x(t) = e^{j\omega_0 t} = \cos(\omega_0 t) + j\sin(\omega_0 t)$



periodicity : $x(t) = x(t + T_0)$

$$e^{j\omega_0 t} = e^{j\omega_0 (t+T_0)}$$

$$\omega_0 T_0 = 2\pi n,\ n = \pm 1, \pm 2 \cdots$$

basic cycle : $T_0 = \dfrac{2\pi}{|\omega_0|}$

It has the same period as sine and cosine signals of the same frequency

3. Exponential signals

- Imaginary exponential signals $x(t) = \mathrm{e}^{\mathrm{j}\omega_0 t} = \cos(\omega_0 t) + \mathrm{j}\sin(\omega_0 t)$

Euler formula

$$\cos(\omega_0 t) = \frac{1}{2}(\mathrm{e}^{\mathrm{j}\omega_0 t} + \mathrm{e}^{-\mathrm{j}\omega_0 t})$$

$$\sin(\omega_0 t) = \frac{1}{2\mathrm{j}}(\mathrm{e}^{\mathrm{j}\omega_0 t} - \mathrm{e}^{-\mathrm{j}\omega_0 t})$$

Euler formula establishes the relationship between positive and cosine signals and virtual exponential signals

Leonhard, Euler

(1707～1783)

Swiss mathematician and physicist, one of the forerunners of modern mathematics.

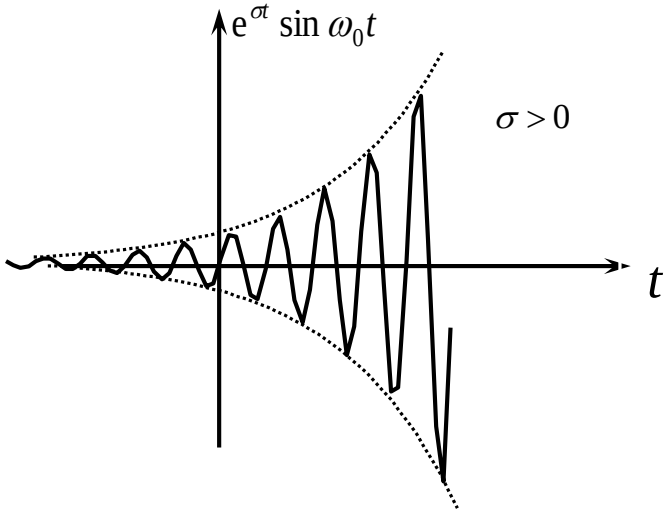
Euler was a prolific mathematician in the history of mathematics, writing an average of more than 800 pages of papers every year, and also writing a large number of textbooks on mechanics, analysis, geometry, variational method and so on, among which "Introduction to Infinitesimal Analysis", "Introduction to Differential Theory", "Integral Theory" and so on have become classic works in mathematics.

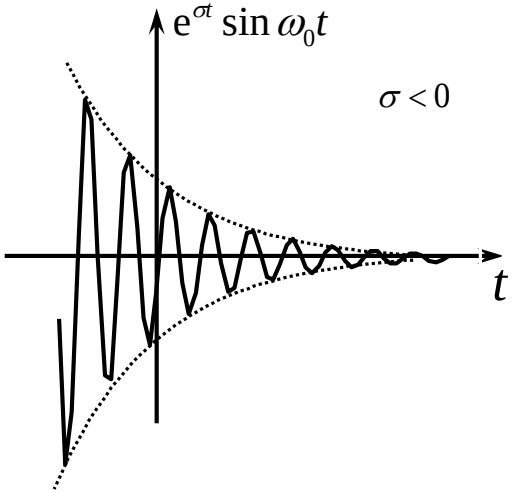
In many branches of mathematics, we often see important formulas and theorems named after Euler, such as Euler function in number theory, Euler formula, and Euler equation in fluid mechanics. The most famous is euler's first use of the concept of "function" to describe expressions containing various parameters, denoted as $f(x)$

3. Exponential signals

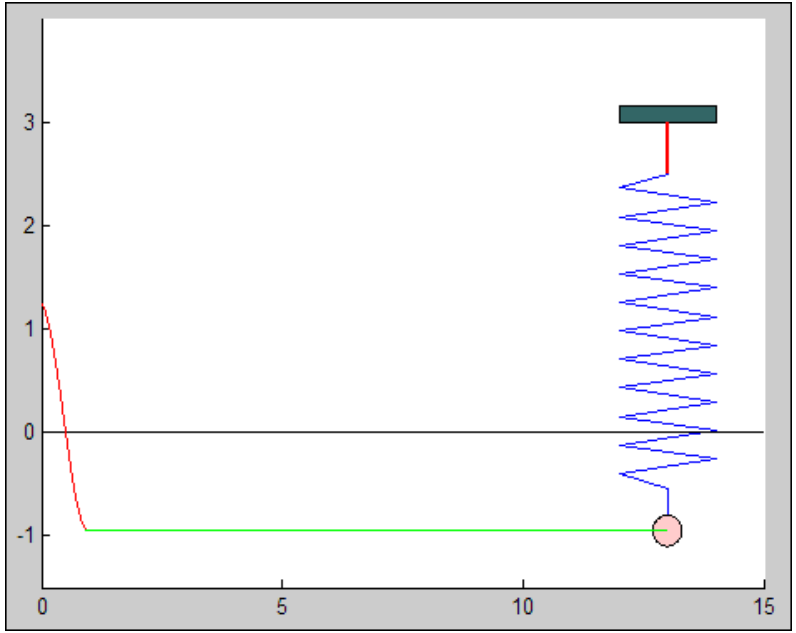
- complex exponential signals $x(t) = Ae^{st}$ $s = \sigma + j\omega_0$

$$x(t) = Ae^{\sigma t}e^{j\omega_0 t} = Ae^{\sigma t}\cos\omega_0 t + jAe^{\sigma t}\sin\omega_0 t$$





3. Exponential signals

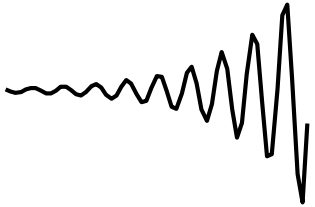


The damping vibration process of spring is simulated

3. Exponential signals

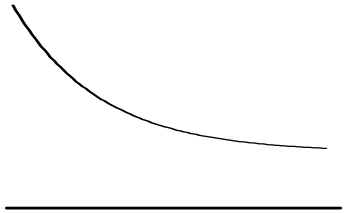
- complex exponential signals $x(t) = Ae^{st} \quad s = \sigma + j\omega_0$

when $\sigma \neq 0$ and $\omega_0 \neq 0$, Unequal oscillation sinusoidal signal



when $s = 0$ and $\omega_0 \neq 0$, imaginary exponential signal

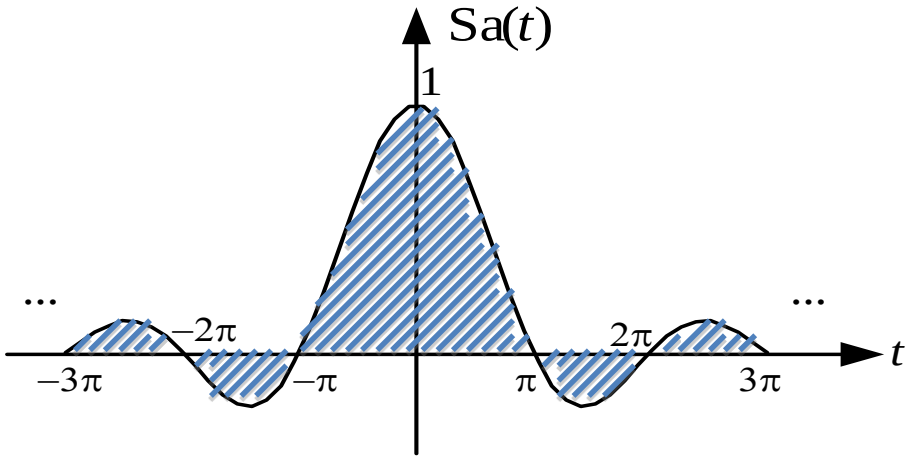
when $\sigma \neq 0$ and $W_0 = 0$, real exponential signal



when $s = 0$ and $W_0 = 0$, DC signal

4. Sampling signals

$$\mathrm{Sa}(t) = \frac{\sin t}{t}$$



character :

$$\mathrm{Sa}(0) = 1$$

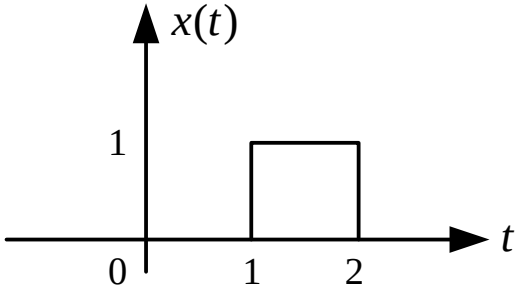
$$\mathrm{Sa}(k\pi) = 0, k = \pm 1, \pm 2 \cdots$$

$$\int_{-\infty}^{\infty} \mathrm{Sa}(t)\mathrm{d}t = \pi$$

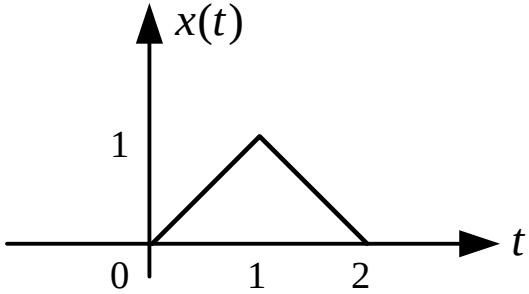
Singularity signals

- definition

The signal $x(t)$ itself or its derivative or higher-order derivative has discontinuous points (jump points). 。



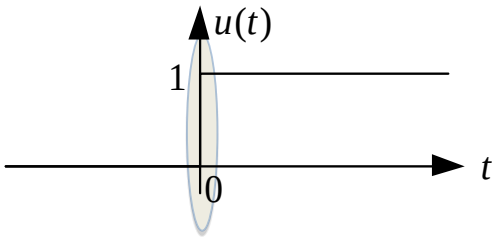
The signal itself has discontinuous points



The derivative of the signal has discontinuous points

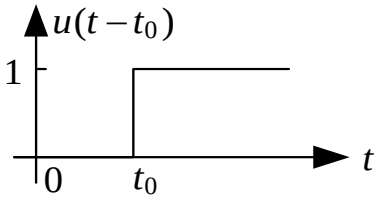
1. Unit step signal

$$u(t)=\begin{cases}1 & t>0\\ 0 & t<0\end{cases}$$

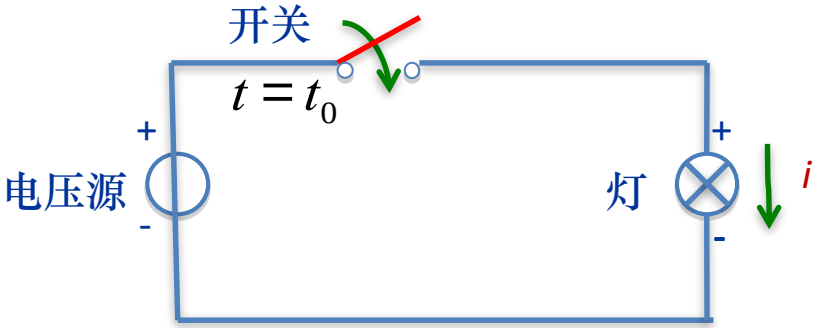


Note:
At the jump point 0, $u(0)$ undefined。

$$u(t-t_0)=\begin{cases}1 & t>t_0\\ 0 & t<t_0\end{cases}$$

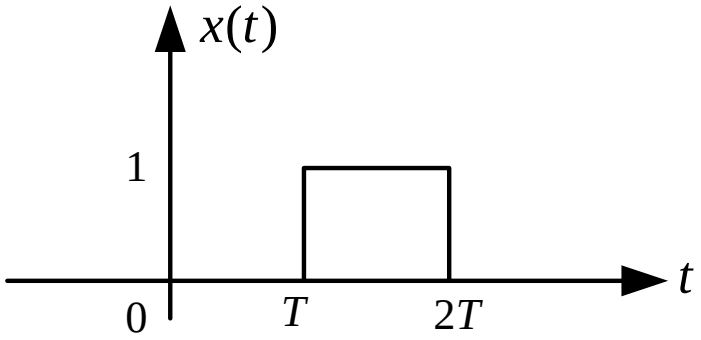


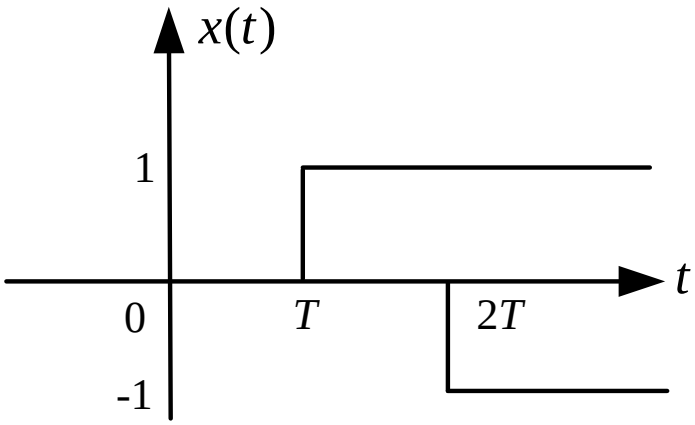
- Practical questions:
- Switching circuits



1. Unit step signal

- function: (1) Indicates an arbitrary rectangular pulse signal

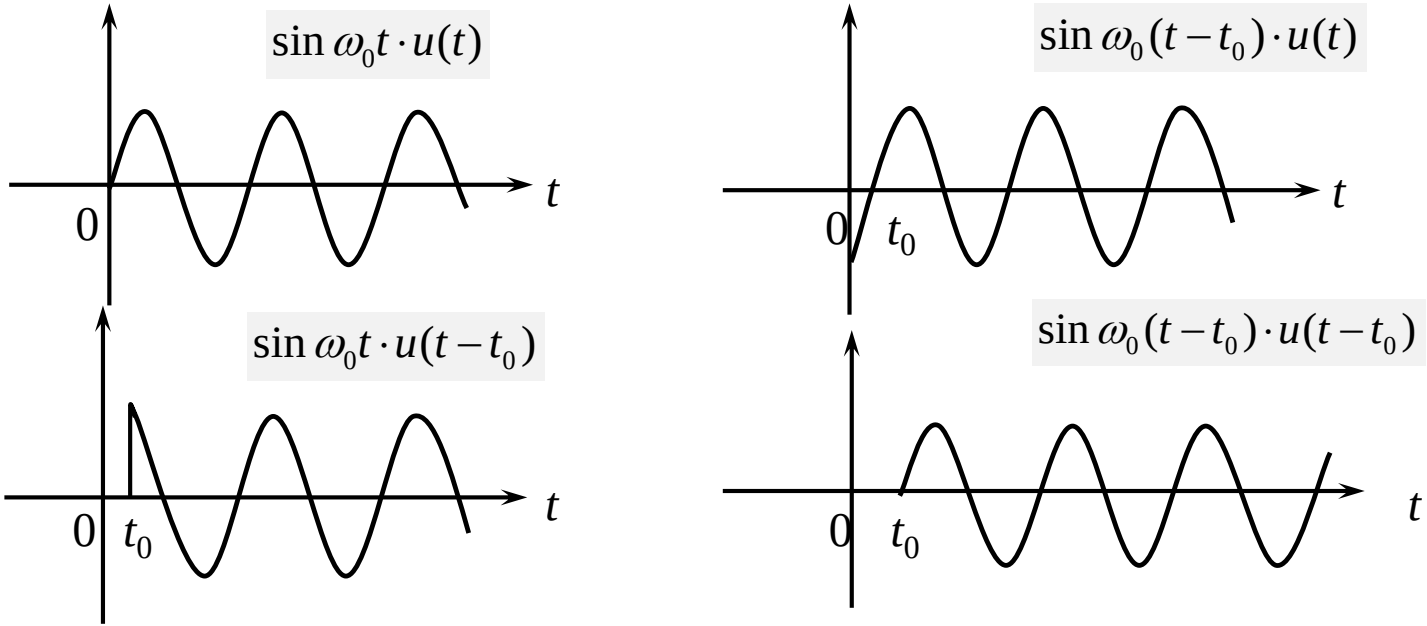




$$x(t) = u(t-T)-u(t-2T)$$

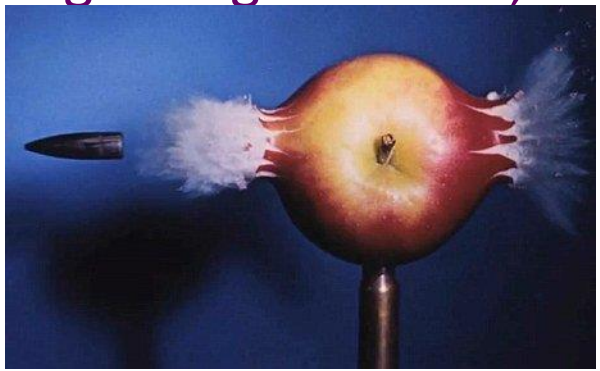
1. Unit step signal

- function: (2) Use the unilaterality of the step signal to express the time range of the signal



2. Unit impulse signal

- Practical problems: instantaneous impact in mechanics, lightning in nature, etc.



Bullets pierce apples

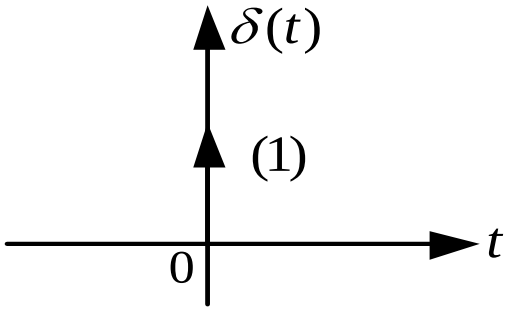
lightning

Impulse signals can well describe these actual phenomena!

Features: Extremely short action time and extremely large amplitude

2. Unit impulse signal

- Dirac definition



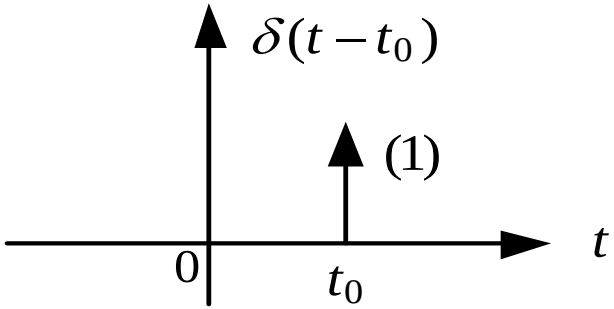
$$\delta(t)=0\ ,\quad t\neq 0$$

$$\int_{-\infty}^{+\infty} \delta(t)\ \mathrm{d}t=1$$

The impulse signal has an intensity, and its strength is the definite integral value of the impulse signal to t. This is noted in parentheses in the figure to distinguish the amplitude of the signal.

2. Unit impulse signal

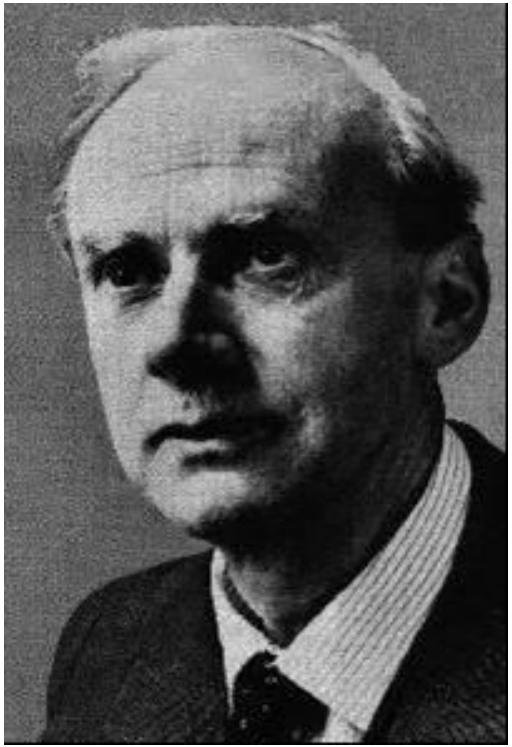
- Dirac definition



$$\delta(t-t_0)=0 \qquad t \neq t_0$$

$$\int_{-\infty}^{\infty} \delta(t-t_0)\mathrm{d}t = \int_{t_0-\Delta}^{t_0+\Delta} \delta(t-t_0)\mathrm{d}t = 1$$

Paul Adrie Maurice Dirac



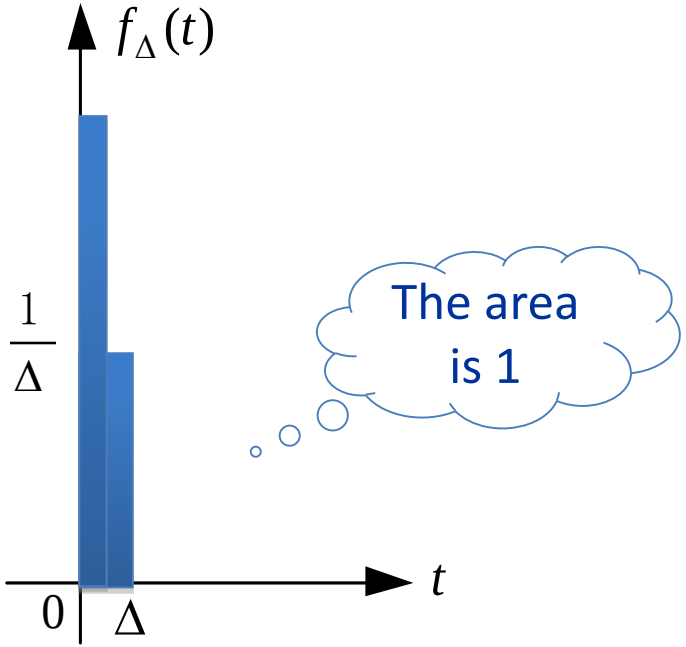
(1902～1984)

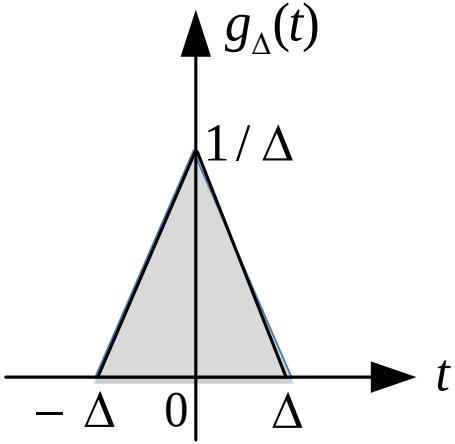
British theoretical physicist, one of the founders of quantum mechanics. In 1935, he was invited by Tsinghua University to give a lecture on positrons at Tsinghua University, and was elected as an honorary member of the Chinese Physical Society.

In 1928 he introduced the theory of relativity to quantum mechanics, establishing the Schrödinger equation in the form of relativity, the famous Dirac equation. Dirac thus made the prediction of the existence of positrons, and was awarded the Nobel Prize in 1933. His major works include Principles of Quantum Mechanics. In addition, he and Fermi independently discovered the Fermi-Dirac method of statistics, published a large number of papers on cosmology, and proposed predictions that magnetic monopoles might exist.

2. Unit impulse signal

- Limit model





$$\delta(t) = \lim_{D\to 0} f_D(t) = \lim_{D\to 0} g_D(t)$$

2. Unit impulse signal

A generalized function definition of the impulse signal $\delta(t)$

$$\int_{-\infty}^{\infty} \phi(t)\delta(t)\mathrm{d}t = \phi(0)$$

$\varphi(t)$ is a test function, which is an arbitrary continuous signal

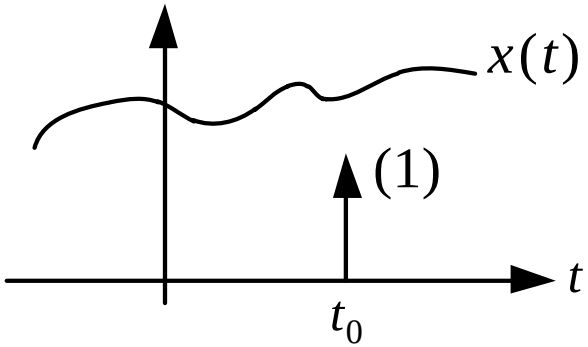
If a signal is integrated with any continuous signal $\varphi(t)$ on (-∞, ∞).

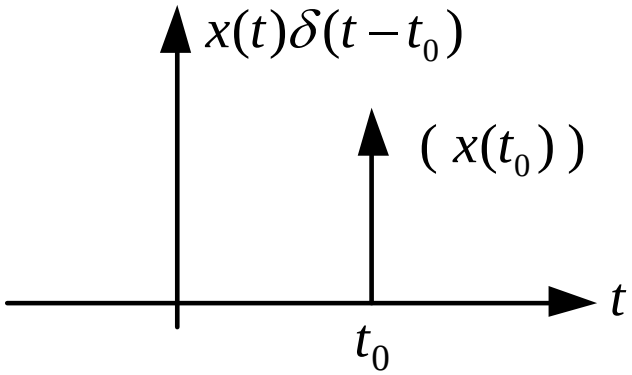
Equal to the value of $\varphi(t)$ at the moment of 0, the signal is an impulse signal.

2. Unit impulse signal

- Property 1: Screening characteristics

$$x(t)\delta(t-t_0) = x(t_0)\delta(t-t_0)$$





It is still an impulse signal,

and the intensity value changes

2. Unit impulse signal

- Property 2: Sampling characteristics

$$\int_{-\infty}^{\infty} x(t)\delta(t-t_0)\mathrm{d}t = x(t_0)$$

Prove: $\int_{-\infty}^{\infty} x(t)\delta(t-t_0)\mathrm{d}t$

$$= \int_{-\infty}^{\infty} x(t_0)\delta(t-t_0)\mathrm{d}t = x(t_0)\int_{-\infty}^{\infty} \delta(t-t_0)\mathrm{d}t = x(t_0)$$

Take advantage of filtering attributes

2. Unit impulse signal

- Property 3: Shrinkage characteristics

$$\delta(at) = \frac{1}{|a|}\delta(t) \qquad (a \neq 0)$$

(1) when $a = -1$, $\delta(t) = \delta(-t)$

Corollary: The impulse signal is an even function

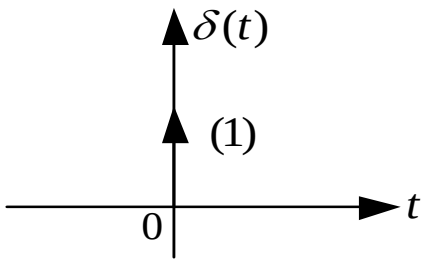
(2) $$\delta(at + b) = \frac{1}{|a|}\delta(t + b/a) \qquad (a \neq 0)$$

This can be proved by generalized function theory

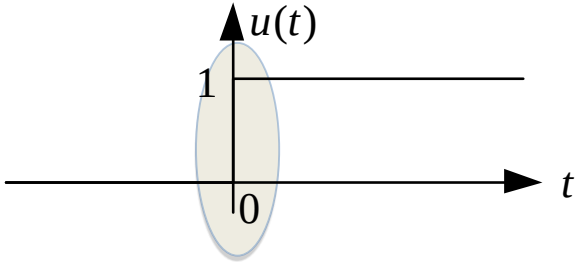
2. Unit impulse signal

- Properties: relationship between impulse signal and step signal

$$\frac{\mathrm{d}u(t)}{\mathrm{d}t} = \delta(t)$$



$$\int_{-\infty}^{t} \delta(\tau)\cdot \mathrm{d}\tau = \begin{cases} 1 & t>0 \\ 0 & t<0 \end{cases} = u(t)$$

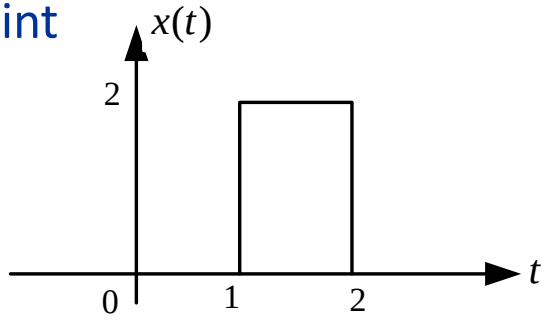


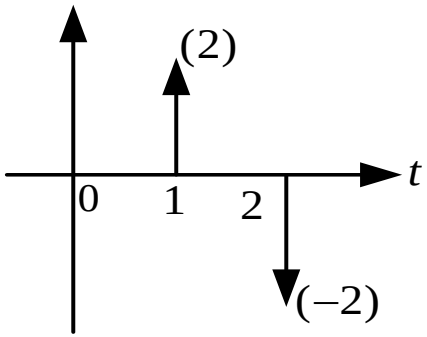
2. Unit impulse signal

$$\frac{\mathrm{d}u(t)}{\mathrm{d}t} = \delta(t)$$

The derivative of the signal at the discontinuous point is the impulse signal or the delayed impulse signal,

The strength of the impulse signal is the jump value at the discontinuous point





[example] Calculate

(1) $\int_{-\infty}^{+\infty} \sin(t) \cdot \delta(t - \frac{\pi}{4}) \mathrm{d}t$

(2) $\int_{-2}^{+3} \mathrm{e}^{-5t} \cdot \delta(t - 1) \mathrm{d}t$

(3) $\int_{-4}^{+6} \mathrm{e}^{-2t} \cdot \delta(t + 8) \mathrm{d}t$

(4) $\int_{-\infty}^{+\infty} \mathrm{e}^{-t} \cdot \delta(2 - 2t) \mathrm{d}t$

(5) $\int_{-2}^{+2} (t^2 + 3t) \cdot \delta(\frac{t}{3} - 1) \mathrm{d}t$

(6) $\mathrm{e}^{-4t} \cdot \delta(2 + 2t)$

[example] Calculate

$$(1)\ \int_{-\infty}^{+\infty}\sin(t)\cdot\delta(t-\frac{\pi}{4})\mathrm{d}t\ =\sin(\frac{\pi}{4})=\sqrt{2}/2$$

$$(2)\ \int_{-2}^{+3}\mathrm{e}^{-5t}\cdot\delta(t-1)\mathrm{d}t\ =\mathrm{e}^{-5\times1}=1/\mathrm{e}^{5}$$

$$(3)\ \int_{-4}^{+6}\mathrm{e}^{-2t}\cdot\delta(t+8)\mathrm{d}t\ \ =0$$

$$(4)\ \int_{-\infty}^{+\infty}\mathrm{e}^{-t}\cdot\delta(2-2t)\mathrm{d}t\ =\int_{-\infty}^{+\infty}\mathrm{e}^{-t}\cdot\frac{1}{2}\delta(t-1)\cdot\mathrm{d}t=\frac{1}{2\mathrm{e}}$$

$$(5)\ \int_{-2}^{+2}(t^2+3t)\cdot\delta(\frac{t}{3}-1)\mathrm{d}t=\int_{-2}^{+2}(t^2+3t)\cdot3\delta(t-3)\cdot\mathrm{d}t=0$$

$$(6)\ \mathrm{e}^{-4t}\cdot\delta(2+2t)\ =\mathrm{e}^{-4t}\cdot\frac{1}{2}\delta(t+1)=\frac{1}{2}\mathrm{e}^{-4\times(-1)}\delta(t+1)=\frac{1}{2}\mathrm{e}^{4}\delta(t+1)$$

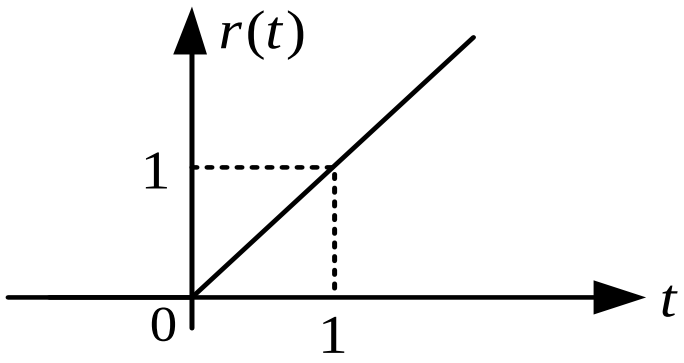
2. Unit impulse signal

1. In the sampling characteristics of the impulse signal, the integral interval may not always be $(-\infty, \infty)$, but as long as the integral interval does not include the $t = t_0$ moment of the impulse signal $\delta(t - t_0)$ the integral result must be zero.

2. For $\delta(at + b)$ impulse signal, the first use of shrinkage characteristics of impulse signal into $\delta\,(t + b/a)/|a|$ form, after can use impulse signal characteristics of sampling and screening

3. Unit ramp signal

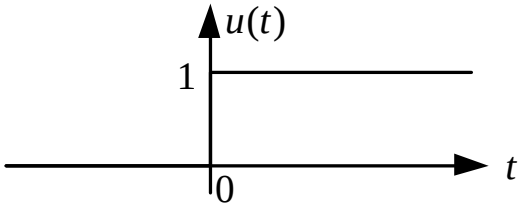
$$r(t)=\begin{cases} t & t\geq 0 \\ 0 & t<0 \end{cases} \Leftrightarrow r(t)=t\cdot u(t)$$



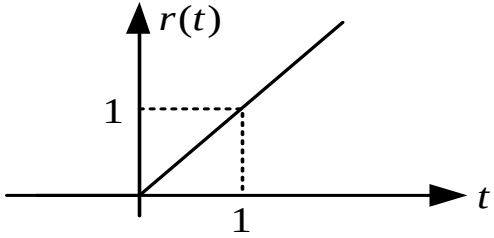
3. Unit ramp signal

- The relationship between the ramp signal and the step signal

$$r(t) = \int_{-\infty}^{t} u(t) \times \mathrm{d}t$$

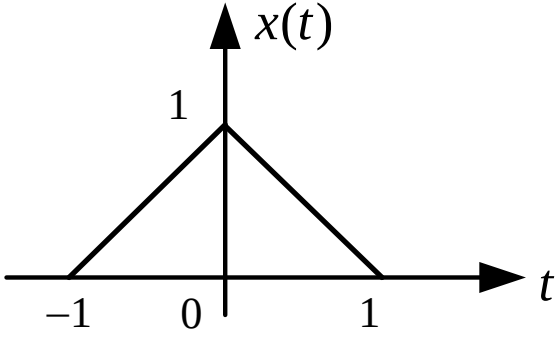


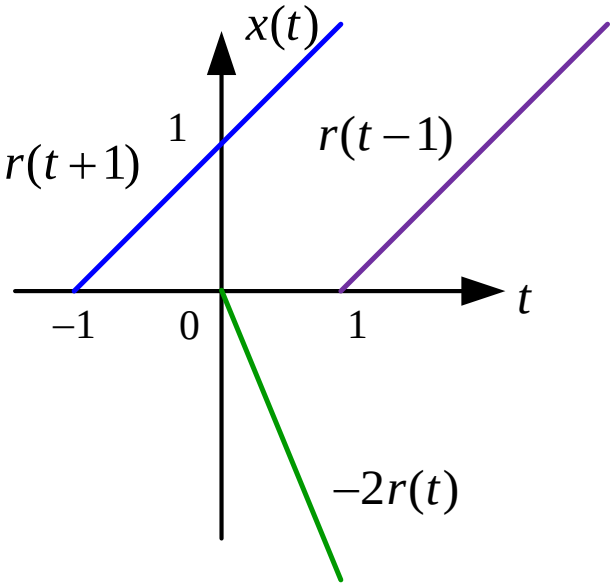
$$\frac{\mathrm{d}r(t)}{\mathrm{d}t} = u(t)$$



3. Unit ramp signal

Represents an arbitrary triangular pulse signal





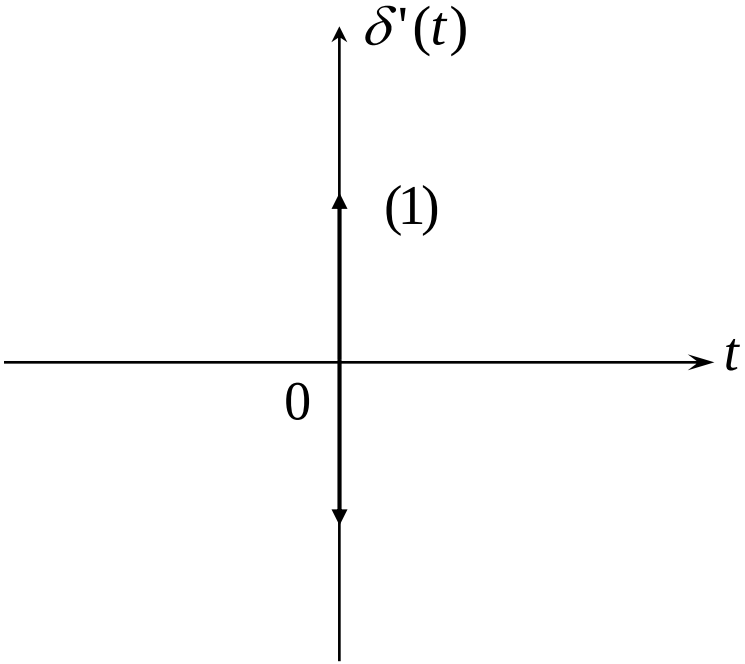
$$x(t) = r(t+1) - 2r(t) + r(t-1)$$

4. Unit impulse pair signal

- **definition:**

$$\delta'(t) = \frac{\mathrm{d}\delta(t)}{\mathrm{d}t}$$

Graphical representation of the impulse pair signal



4. Unit impulse pair signal

Properties of the impulse pair signal:

$$x(t)\delta'(t-t_0)=x(t_0)\delta'(t-t_0)-x'(t_0)\delta(t-t_0)$$

(Screening characteristics)

$$\int_{-\infty}^{\infty} x(t)\delta'(t-t_0)\mathrm{d}t=-x'(t_0)$$

(Sampling characteristics)

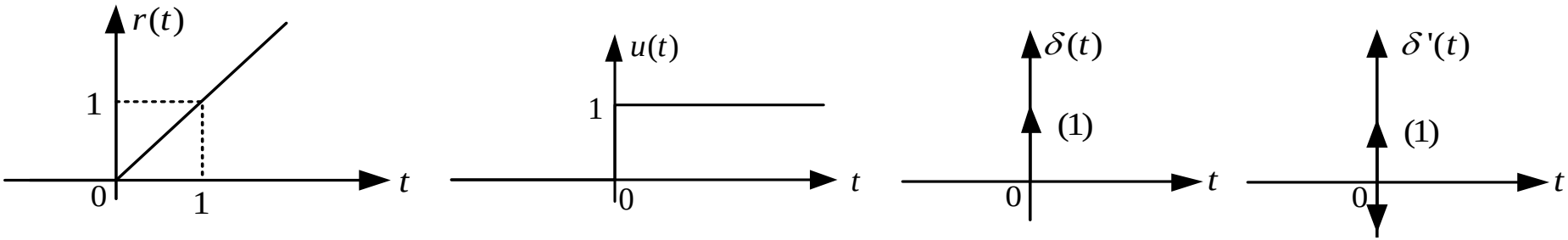
$$\delta'(at)=\frac{1}{a|a|}\delta'(t) \qquad (a\neq 0)$$

(Shrinkage characteristics)

$$\delta'(t)=-\delta'(-t)$$

$$\int_{-\infty}^{\infty}\delta'(t)\mathrm{d}t=0$$

The relationship between four singular signals



Derivative relations $r(t) \xrightarrow{\frac{\mathrm{d}()}{\mathrm{d}t}} u(t) \xrightarrow{\frac{\mathrm{d}()}{\mathrm{d}t}} \delta(t) \xrightarrow{\frac{\mathrm{d}()}{\mathrm{d}t}} \delta'(t)$

Integral relationship $\delta'(t) \xrightarrow{\int_{-\infty}^{t}()\mathrm{d}\tau} \delta(t) \xrightarrow{\int_{-\infty}^{t}()\mathrm{d}\tau} u(t) \xrightarrow{\int_{-\infty}^{t}()\mathrm{d}\tau} r(t)$

Thanks

Some of the materials cited in this course are the teaching accumulation of the main lecturer for many years, which come from a variety of media and the exchanges of colleagues, peers and friends, and it is difficult to indicate the source one by one, so I hereby explain and thank you!